

AD7190

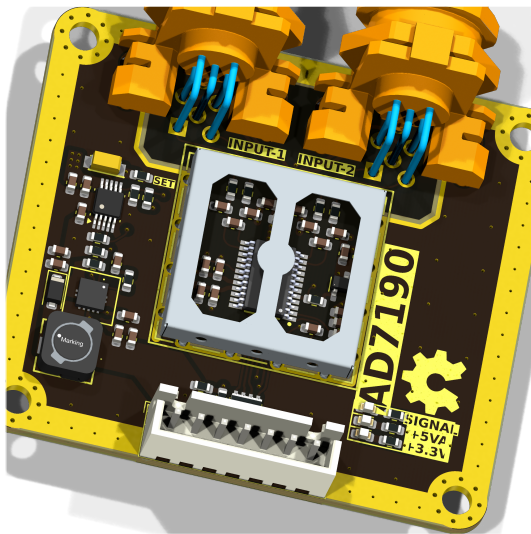
Variant:

2026-10-09

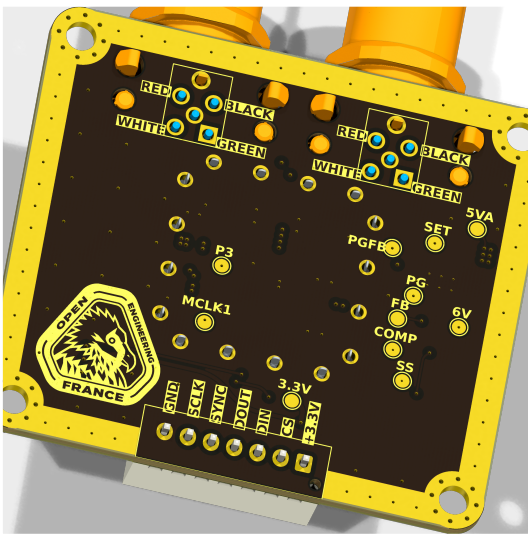
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 09-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
+3.3V host ->
TPS61086
6 V -> LT3042 +5VA
(AVDD + excitation)

DEBUG NOTE:
TP on every rail.
Wait >= 600 ms (PG)
before ADC init.

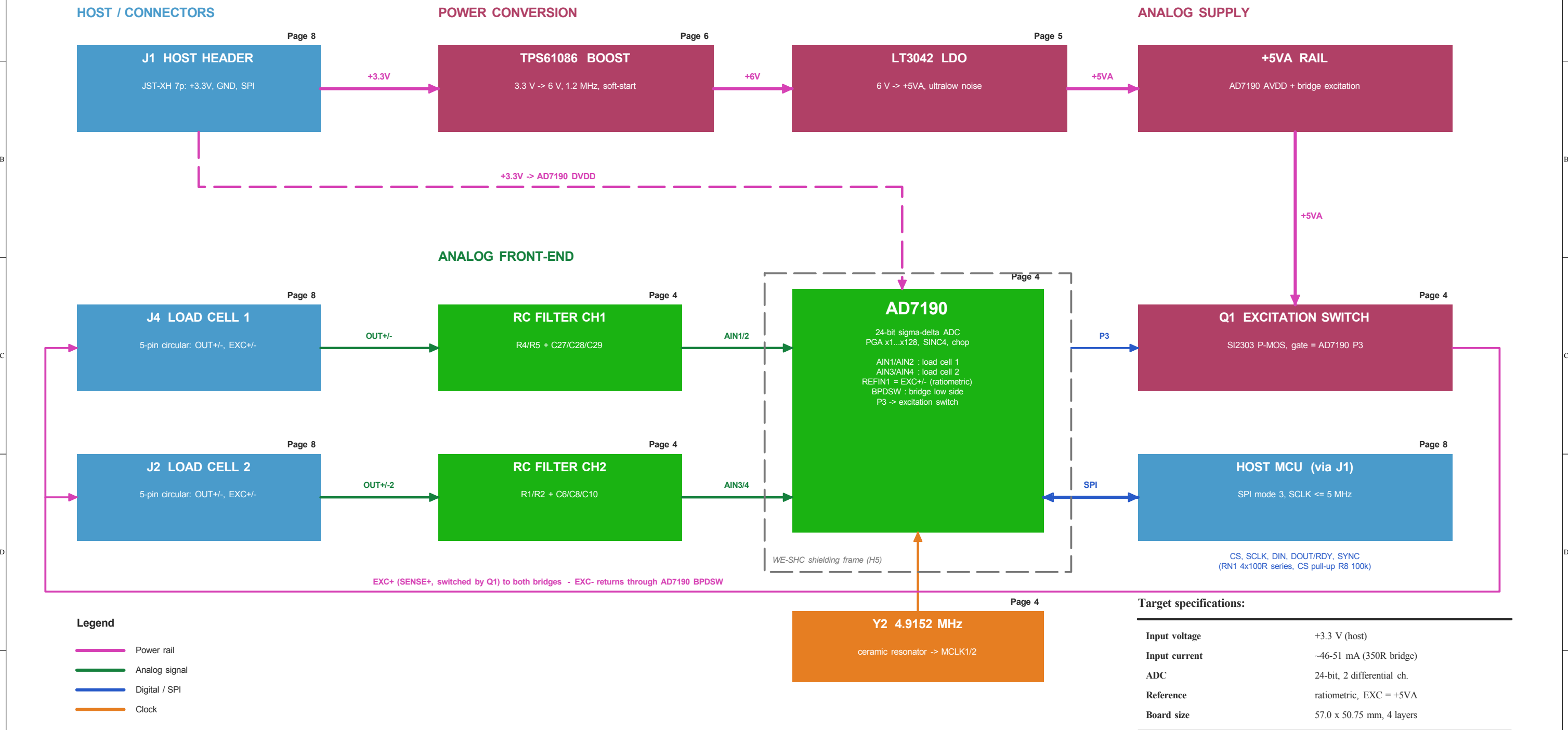
DESIGN NOTE:
SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE:
REFIN1 = bridge
excitation:
ratiometric.

LAYOUT NOTE:
AFE under the
WE-SHC shield H5,
boost SW node
away.

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Title: OPEN_WEIGHT_AD7190	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 1 of 10

[2] Block Diagram



	Comments:		Company:		Variant:		Git Hash:		
			OPEN THRUST LAB				b3867f9		
			Board Name:			Project Name:			
			AD7190			OPEN_WEIGHT			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Block Diagram		Block Diagram.kicad_sch		SIEBERT DIMITRY		2026-09-29		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
/Block Diagram/						A3		2 of 10	

[3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.

HOST INTERFACE

Connectors

J1 host header 7-pin: +3.3V, GND, SPI
J2 / J4 load cells 1 & 2
5-pin circular: OUT±, EXC±
H5 WE-SHC shielding frame (AD7190)

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POWER CONVERSION

TPS61086_3.3V_to_6V

Boost converter +3.3V -> +6V
IC2 TPS61086, 1.2 MHz, soft-start
L1 + Schottky D3, +3.3V LED D1
Test points TP14-TP16

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LT3042_6V_to_5VA

Ultralow-noise LDO +6V -> +5VA
U2 LT3042, SET = 100 µA x 50 kΩ
Tantalum C14 on SET, +5VA LED D4
Test points TP10-TP13, TP17

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ANALOG FRONT-END

AD7190_Weight_Sensor

U1 AD7190 24-bit sigma-delta ADC
2 differential ch., RC input filters
Ratiometric ref, REFIN1 = EXC±
Y2 4.9152 MHz, Q1 excitation switch
Test points TP1-TP9, TP18

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Holes, Fiducials

H1-H4 mounting holes
FID1-FID6 fiducials
No electrical function

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LEGEND

- Power rail
- Secondary supply
- Analog signal
- Digital / SPI

DOCUMENT INDEX

- [1] Title Page

[2] Block Diagram

[3] Project Architecture

[4] AD7190 Weight Sensor

[5] LT3042 LDO 6V -> 5VA
- [6] TPS61086 Boost 3V3 -> 6V

[7] Holes, Fiducials

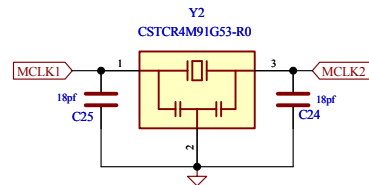
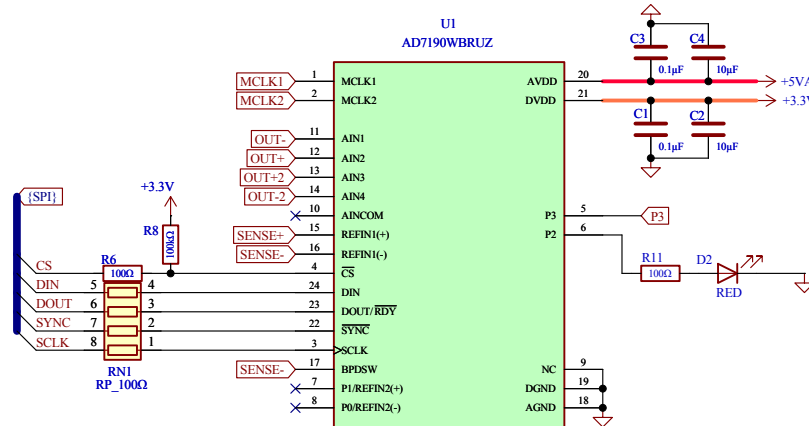
[8] Connectors

[9] Power Sequencing

[10] Revision History

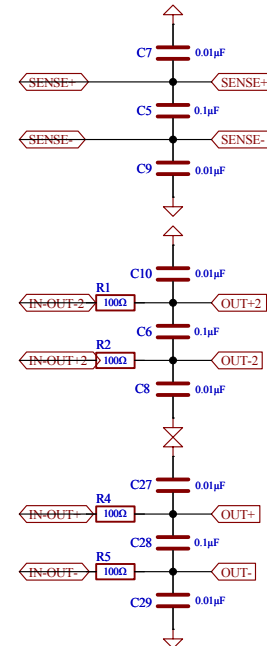
	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 10

[4] AD7190 - Weight Sensor



DESIGN NOTE:
Ref Condo for quartz
Murata GRM1885C1H220JA01D
(22 pF, 50 V, C0G, 0603, ±5%)

ADC FOR
weighing sensor



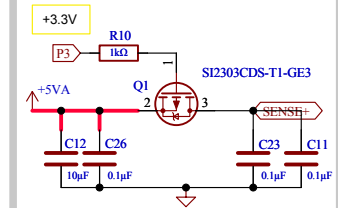
Calcul de filtre : $f_c = 1 / (2\pi \times R \times C) = X \text{ Hz}$

DESIGN NOTE:

Ref Condo for Filtre
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

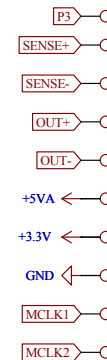
Ref Resistor for Filtre
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

FILTRE



DESIGN NOTE:
IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation. must disabled during calibration

DRIVE GATE



TEST POINT

About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta ($\Sigma\Delta$) analog-to-digital converter (ADC). The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC. The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz. The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.



Comments:
AD7190
CSTCR4M91G53-R0
AD7190_TEST_BOARD CN

Sheet Title:
AD7190 - Weight Sensor

Sheet Path:
/Project Architecture/AD7190_Weight_Sensor/

Company:
OPEN THRUST LAB

Board Name:
AD7190

File Name:
AD7190 - Weight Sensor.kicad_sch

Designer:
SIEBERT DIMITRY

Reviewer:

Variant:

Git Hash:
b3867f9

Project Name:
OPEN_WEIGHT

Date:
2026-09-29

Revision:
+ (Unreleased)

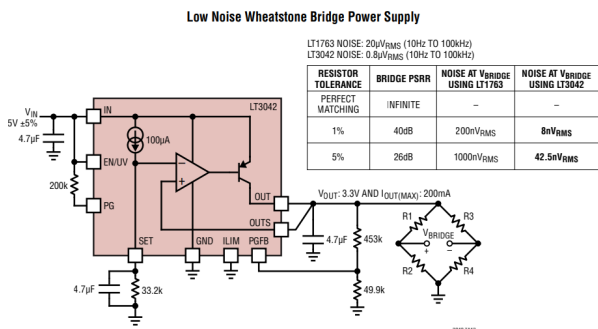
Size:
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Sheet:
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[5] LT3042 - LDO

6V to 5V Analog

TYPICAL APPLICATIONS



IN (pins 1-2): LT3042 power input (6V), to be decoupled with CIN.

EN/UV (pin 3): enables/disables the regulator and allows an UVLO threshold (often tied to IN for "always ON").

PG (pin 4): open-collector Power-Good output (requires a pull-up resistor).

ILIM (pin 5): current limit programming/monitoring (often tied to GND if unused).

PGFB (pin 6): input that sets the Power-Good threshold via a resistor divider from VOUT.

SET (pin 7): sets the output voltage via RSET (and CSET reduces noise / provides soft-start).

GND (pin 8): circuit ground reference.

OUTS (pin 9): Kelvin output sense (to be connected to the + of COUT / load point).

OUT (pin 10): regulated output (5V), to be decoupled with COUT.

EPAD (exposed pad): ground + thermal dissipation (to be soldered to the GND plane).

DESIGN NOTE:

Ref Condo for SET PIN (important pour les vibrations P16)
KYOCERA AVX TAJA225K016RNJ (2.2 µF, 16 VDC,
condensateur tantale solide SMD moulé, 1206 / 3216 (A case),
±10%, ESR 6.5 Ω).

$V_{OUT} \approx I_{SET} \times R_{SET}$
 $I_{SET} \approx 100 \mu A$

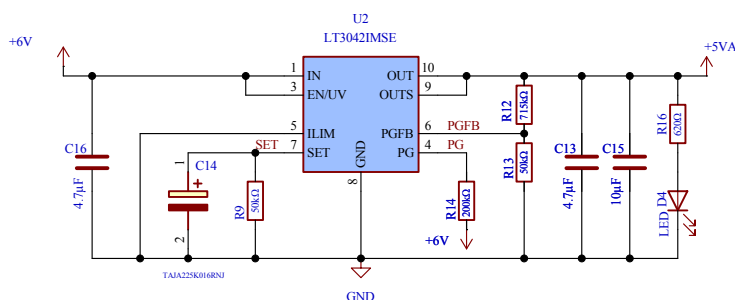
Donc :
 $R_{SET} \approx V_{OUT} / I_{SET}$
 $R_{SET} \approx 5.0 V / 100 \mu A \approx 50 k\Omega$
(=> valeur standard recommandée : 49.9 kΩ)

$V_{OUT}(PG_THRESHOLD) \approx 0.3 V \times (1 + R_{PG2}/R_{PG1})$
 $PGFB_THRESHOLD \approx 0.3 V$

Donc :
 $R_{PG2} \approx R_{PG1} \times (V_{OUT}(PG)/0.3 V - 1)$

Exemple (PG OK à ~4.8 V) :
Choix $R_{PG1} = 49.9 k\Omega$
 $R_{PG2} \approx 49.9 k\Omega \times (4.8/0.3 - 1) \approx 748.5 k\Omega$
(=> valeur standard recommandée : 750 kΩ)

Design guide line



Convertisseur 5VA

About

The LT®3042 is a high performance low dropout linear regulator featuring LTC's ultralow noise and ultrahigh PSRR architecture for powering noise sensitive RF applications. Designed as a precision current reference followed by a high performance voltage buffer, the LT3042 can be easily paralleled to further reduce noise, increase output current and spread heat on the PCB. The device supplies 200mA at a typical 350mV dropout voltage. Operating quiescent current is nominally 2mA and drops to <<1µA in shutdown. The LT3042's wide output voltage range (0V to 15V) while maintaining unitygain operation provides virtually constant output noise, PSRR, bandwidth and load regulation, independent of the programmed output voltage. Additionally, the regulator features programmable current limit, fast start-up capability and programmable power good to indicate output voltage regulation. The LT3042 is stable with a minimum 4.7µF ceramic output capacitor. Built-in protection includes reverse battery protection, reverse current protection, internal current limit with foldback and thermal limit with hysteresis. The LT3042 is available in thermally enhanced 10-Lead MSOP and 3mm × 3mm DFN package

TEST POINT



Comments:

LT3042
Analog Devices

Sheet Title:

LT3042 - LDO 6V to 5V Analog

Sheet Path:

/Project Architecture/LT3042_6V_to_5VA/

Company:

OPEN THRUST LAB

Board Name:

AD7190

File Name:

LT3042 - LDO 6V to 5VA.kicad_sch

Designer:

SIEBERT DIMITRY

Reviewer:

Variant:

Project Name:

OPEN_WEIGHT

Date:

2026-09-29

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Git Hash:

b3867f9

Revision:

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Revision:

(Unreleased)

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The schematic shows a buck converter circuit centered around the TPS61086 IC. The input voltage (VIN) is connected to the SW pin (pin 7). A series resistor is placed between VIN and the SW pin. The output of the converter (VOUT) is taken from the OUT pin (pin 9), which is also connected to a load resistor. The ground connection (GND) is shared by several pins: SS (pin 10), IN (pin 8), SW (pin 6), FB (pin 2), GND (pin 4), and POND (pin 5). The COMP pin (pin 1) is connected to a compensation network consisting of a capacitor and a resistor. The MODE pin (pin 9) is connected to a pull-up resistor. The SW pin (pin 7) is connected to a switching transistor (MOSFET) through a gate driver stage.

Pin usage and layout example



Test points

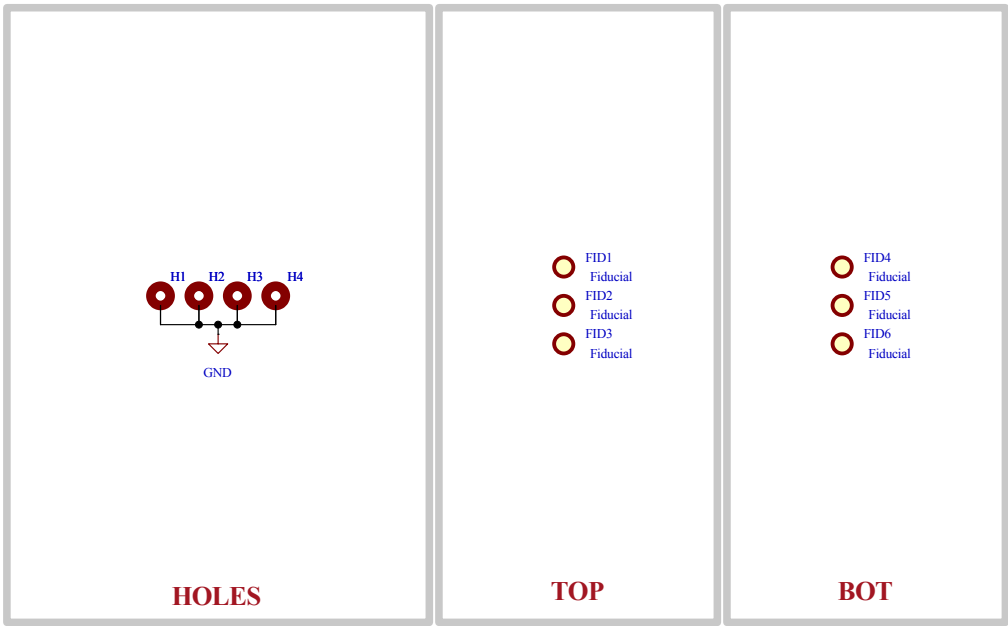
At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.



Sheet:

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[7] Holes, Fiducials



	Comments: References: 4 x M3 (3.2 mm) plated holes to GND FID1-FID6 not fitted on this revision	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Sheet Title: Holes, Fiducials	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /Project Architecture/Holes, Fiducials/	File Name: Holes - Fiducials.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 7 of 10

[8] Connectors

DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

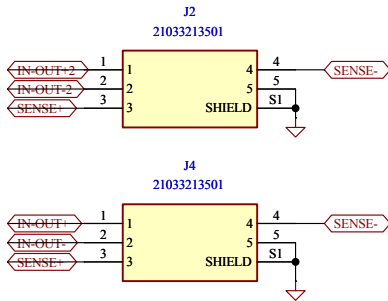
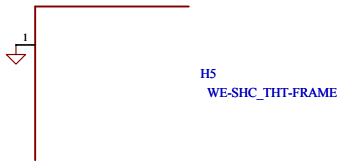
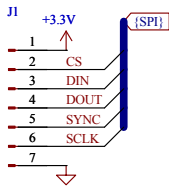
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,
J2/J4 21033213501, H5 WE-SHC



Host and load-cell connectors



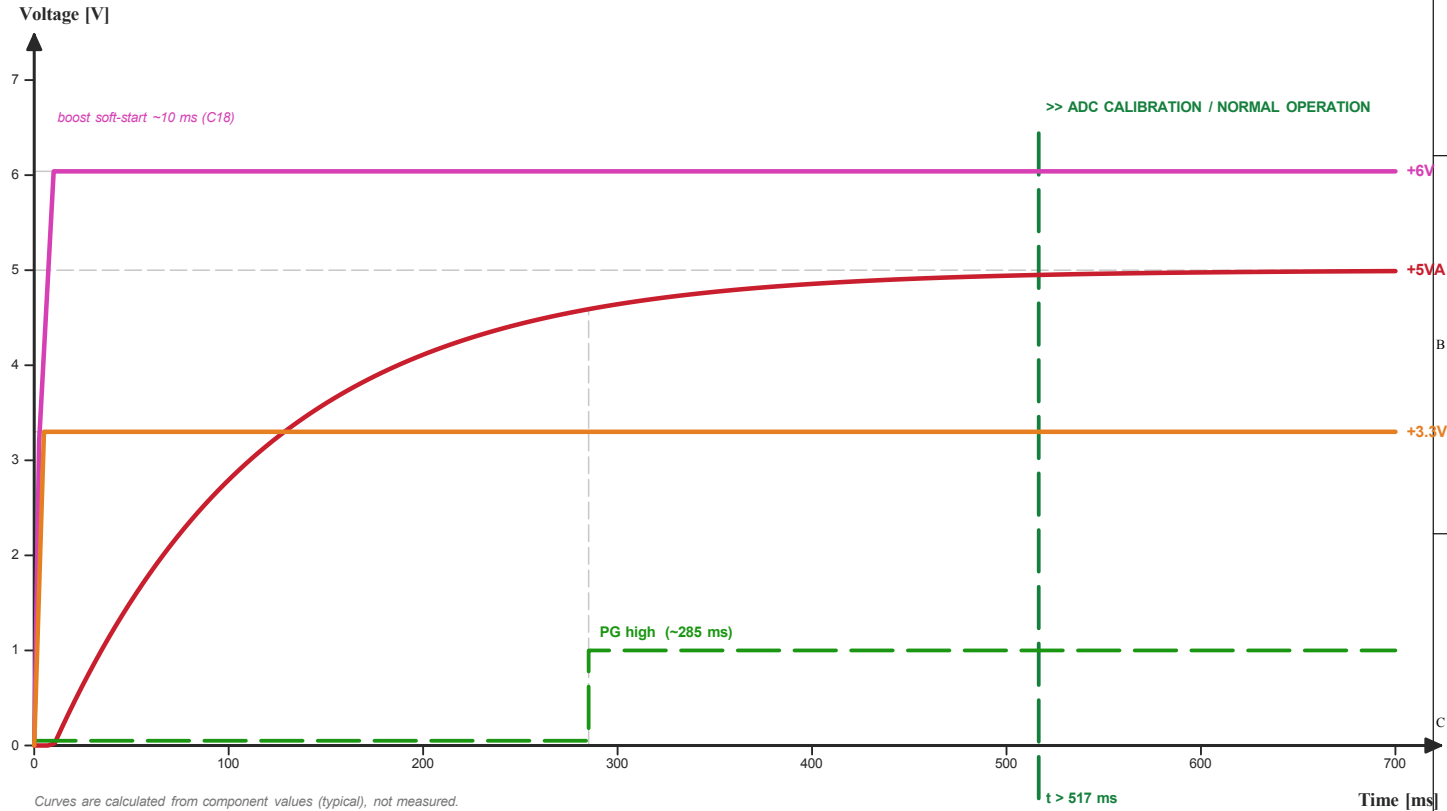
Comments: References: JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A3	Revision: + (Unreleased)
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[9] Power - Sequencing

NAME	SOURCE	LEVEL
+3.3V	Host (J1)	3.3 V (host)
+6V	TPS61086 IC2	6.04 V
+5VA	LT3042 U2	5.00 V
PG	LT3042 U2	> 4.59 V

Values from the schematic:
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)
+5VA = ISET x RSET = 100 uA x 50k (R9)
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
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	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

Version Unreleased

Added

- AD7190 front-end, 2 load-cell inputs
- +3.3V -> TPS61086 6 V -> LT3042 +5VA
- Self-contained libs and 3D models (lib/)
- JLCPCB 4-layer rules and stackup
- LTspice simulations (Simulation/LTspice)
- openEMS layout coupling (Simulation/openEMS)

Changed

- Renamed to OPEN_WEIGHT, KiCad 10

Fixed

- Connectors not linked to ADC in schematic
- AD7190 solder-mask bridges

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Revision History	File Name: Revision_History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10